Equilibrium Atlases for Biochemical ODE Models: Conditional Certificates and an Implementation Audit

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Abstract

We study a tool for counting and classifying equilibria uniformly over parameter cells using established interval Krawczyk and matrix-inertia methods. The mathematical certificate is conditional on valid interval enclosures and complete enumeration. An implementation audit found and repaired a false zero-root certificate caused by overflow and invalid interval endpoints; finite tests cannot establish universal implementation soundness. In a prospectively fixed comparison on 18 affine and fold cells, evaluated with scalar and vector backends, 22 of 36 runs produce analytically correct certificates and 14 remain unknown. A direct analytic solver resolves all 30 runs whose cells admit a uniform hyperbolic answer; the other six cross a fold or contain its nonhyperbolic point. Fresh extended-bound composition tests enclose 7,009 exact rational finite-member answers, with 991 domain exceptions retained. Earlier archives report 11,035 certified cells among 22,173 cells across 21 models, but were not rerun after the arithmetic repairs. This paper therefore presents an audited implementation checkpoint and a reproducible historical archive, rather than a new validated-solving theorem or demonstrated superiority over mature complete solvers.

1 Introduction

Qualitative questions about a parametrised biochemical model, such as *for which parameters is the switch bistable* or *where does the steady state lose stability*, are answered in practice with continuation software [6] or by locating equilibria with Newton's method on a parameter grid. These answers are reliable at the computed points. They do not certify the absence of further equilibria, and they extrapolate between grid points. When the answer feeds a design decision, such as choosing parameters inside the bistable region, or a model-selection argument, such as showing that a model cannot be bistable for physiological parameters, a statement that holds for whole parameter regions is more useful than a dense set of point answers.

Claim. The intended output is an equilibrium count and inertia signature valid throughout each certified parameter cell. The mathematical argument assumes valid enclosures and completed enumeration. Current finite tests close identified arithmetic counterexamples; they do not prove that every implementation path satisfies those assumptions. The supported contribution is the implementation, its documented failure boundary, and reproducible evidence, with historical atlas results explicitly separated from fresh post-repair tests.

Contributions.

1. A *cell certificate* (Definition 1) and an algorithm that produces it by complete enumeration of equilibria by branch and prune with a parametric Krawczyk operator (Section 3.2), and an inertia certificate based on the Ostrowski–Schneider theorem whose positivity test is an interval Cholesky decomposition (Section 3.3).

2. An implementation for models given as SymPy expressions, YAML files or SBML, with automatic conservation-law reduction, vectorised directed-rounding interval arithmetic and a command-line atlas builder (Section 4).

3. An evaluation on 21 models, three of them BioModels entries, plus two larger BioModels entries for scaling (Section 5). It measures certified coverage and cost, checks certificates against floating-point sampling and by independent re-verification with CAPD [15], Z3 [5, 14] and dReal [8], and compares the certificates with the switching-limit analysis of DSGRN [4] and with heuristic stability labels; scaling is measured up to 20 state variables.

2 Background and related work

Validated numerics. Interval Newton and Krawczyk operators [18, 21, 22] prove existence and uniqueness of zeros in boxes; combined with exclusion tests and bisection they give complete solvers for nonlinear systems [16]. Contractor-based solvers such as Ibex [3] follow the same principle. IbexSolve also provides validated underconstrained pavings with existence and uniqueness uniformly over interior values of selected coordinates [13]. Thus parameter-uniform zero certification itself is established prior art; organising it with inertia into a biochemical atlas requires a task-matched comparison before a distinct methodological advance can be claimed. Rump [24] surveys verification methods based on floating-point arithmetic, and CAPD::DynSys [15] is a mature C++ library for rigorous computation in dynamical systems. Rigorous numerics has been used for many computer-assisted proofs in dynamics [26]. PhaseCartographer applies these methods to one task, an exhaustive count of equilibria that is uniform over a parameter box together with a stability classification of the same strength, organised as an atlas.

SMT solvers over the reals. Z3 decides nonlinear real arithmetic with the nlsat procedure [14]; dReal [8] answers δ-complete queries for nonlinear real functions, and its `unsat` answers are proofs. Both can verify individual claims ("no equilibrium outside these boxes", "the Routh–Hurwitz conditions hold on this box") but do not by themselves produce the boxes or certify existence uniformly in the parameters. We use them as independent checkers.

Combinatorial dynamics. DSGRN [4] computes Morse graphs for every region of a combinatorial parameter graph of a regulatory network in the switching (infinitely steep) limit. Its answers are exact for switching systems and are known to transfer to steep Hill functions [7]. For moderate Hill coefficients the dynamics can differ, which our comparison illustrates.

Continuation. Numerical continuation (e.g. MATCONT [6]) traces equilibrium branches and detects bifurcations efficiently, and is the standard tool for bifurcation analysis. It does not certify that all branches have been found.

3 Approach

We consider $\dot{x} = f(x, p)$ with $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ given symbolically, a state box $D \subset \mathbb{R}^n$, optional admissibility constraints $g_j(x, p) \geq 0$ (used for conservation laws, Section 4), and a parameter box $P_0 \subset \mathbb{R}^m$. Intervals and interval vectors are written $[x]$; $F([x], [p])$ and $J([x], [p])$ denote rigorous interval enclosures of the ranges of f and of $\partial f/\partial x$.

Definition 1 (Cell certificate)**.** *For a parameter box $P \subseteq P_0$, a cell certificate consists of boxes $E_1, \ldots, E_k \subseteq D$ and integers $u_1, \ldots, u_k$ such that, for every $p \in P$: (i) $f(\cdot, p)$ has exactly k zeros in the admissible part of D, one in each E_i; (ii) each zero $x_i(p)$ is hyperbolic and $J(x_i(p), p)$ has exactly u_i eigenvalues with positive real part.*

The multiset $\{u_i\}$ is the cell's *signature*; the number of stable equilibria ($u_i = 0$) gives the regime label (monostable, bistable, ...). Statements about periodic orbits are not part of the certificate; a cell whose unique equilibrium is unstable is labelled *unique unstable* and nothing more.

3.1 Parametric Krawczyk operator

For a box B, a point $m \in B$, $p_0 \in P$ and any real matrix R, define

$$K(B, P) = m - R\,\Phi + \big(I - R\,J(B, P)\big)(B - m), \tag{1}$$

where Φ encloses $\{f(m, p) : p \in P\}$. We use the intersection of the natural enclosure $F(m, P)$ with the mean-value form $F(m, p_0) + F_p(m, P)(P - p_0)$, with R multiplied into F_p before the parameter offsets so that cancellations between components are kept.

Theorem 2 (Krawczyk; see [22], Ch. 5)**.** *Fix $p \in P$. Every zero of $f(\cdot, p)$ in B lies in $K(B, P)$. If $K(B, P) \subseteq \operatorname{int} B$, then $f(\cdot, p)$ has exactly one zero in B.*

Since Φ and $J(B, P)$ enclose the values for every $p \in P$, the operator for the fixed p is contained in $K(B, P)$, so both conclusions hold uniformly over P. In particular $K(B, P) \cap B = \emptyset$ proves that B contains no zero for any $p \in P$.

3.2 Complete enumeration

Algorithm 1 enumerates all zeros in D for a cell P. Floating-point Newton iterations from random starts at several parameter points propose zeros z; for each, boxes of increasing radius around z are tested with (1) and the first box Z with $K(Z, P) \subseteq \operatorname{int} Z$ is kept as a *uniqueness box*, together with the enclosure $E = K(Z, P)$. Seeds only save work; nothing about them is trusted. The state box is then processed by branch and prune.

Proposition 3. *If Algorithm 1 returns enclosures $E_1, \ldots, E_k$, then for every $p \in P$, $f(\cdot, p)$ has exactly k zeros in the admissible part of D and the i-th lies in E_i.*

Proof sketch. Every tile that is discarded contains no admissible zero (constraint or range exclusion, $K \cap T = \emptyset$, or an empty Gauss–Seidel contraction, which is a mean-value argument), or contains only the unique zero of a uniqueness box. Contractions keep all zeros of the tile (Theorem 2). Hence every admissible zero lies in some uniqueness box. Two pairs with $E_i \subseteq Z_j$ describe the same zero because Z_j contains only one; pairs with disjoint enclosures describe different zeros. Each uniqueness box contains exactly one zero for each p. □

Algorithm 1 Certified enumeration of equilibria in D for all $p \in P$

$\mathcal{U} \leftarrow$ verified (uniqueness box, enclosure) pairs from Newton seeds
$\mathcal{S} \leftarrow \{D\}$
while $\mathcal{S} \neq \emptyset$ **do**
  take T from $\mathcal{S}$
  if $T \subseteq Z$ for some $(Z, E) \in \mathcal{U}$ **then continue**
  end if
  if $g_j(T, P) < 0$ for some j, or $0 \notin F_i(T, P)$ for some i **then continue**
  end if
  $K \leftarrow K(T, P)$
  if $K \cap T = \emptyset$ **then continue**
  end if
  if $K \subseteq \operatorname{int} T$ **then** add (T, K) to $\mathcal{U}$; **continue**
  end if
  $T \leftarrow$ Gauss–Seidel contraction of $K \cap T$
  if $T = \emptyset$ **then continue**
  end if
  if budget exhausted or T too small **then return** INCOMPLETE
  end if
  bisect T (smear heuristic [16]) and push both halves
end while
merge pairs: $E_i \subseteq Z_j \Rightarrow$ same zero; $E_i \cap E_j = \emptyset \Rightarrow$ distinct; otherwise **return** INCOMPLETE
return the merged enclosures that lie in the admissible part of D

3.3 Inertia certificate

Let $[J] = J(E, P)$ for an enclosure E of the zero $x(p)$. Then $J(x(p), p) \in [J]$ for every $p \in P$.

Theorem 4 (Ostrowski–Schneider [23])**.** *If a real symmetric H satisfies $AH + HA^\top \succ 0$, then A has no eigenvalue on the imaginary axis, and the number of eigenvalues of A with positive (negative) real part equals the number of positive (negative) eigenvalues of H.*

We take $A_0 = \operatorname{mid}[J]$, solve $A_0 H + H A_0^\top = I$ in floating point (Bartels–Stewart [2]), symmetrise H, and then prove two facts rigorously.

Step 1: $AH + HA^\top \succ 0$ for all $A \in [J]$. We enclose $[S] = [J]H + H[J]^\top$ entrywise, intersect $[S]_{ij}$ with $[S]_{ji}$, and run the Cholesky algorithm in interval arithmetic [1]. If every radicand has a positive lower bound, then for each symmetric $S \in [S]$ the point Cholesky algorithm produces quantities contained in the interval ones (inclusion monotonicity), so all its pivots are positive and S is positive definite.

Step 2: the inertia of H. With Q from a floating-point eigendecomposition of H we enclose $M = Q^\top H Q$. If every Gershgorin disc of M avoids 0, then M is nonsingular (hence so is Q), the numbers of positive and negative eigenvalues of M equal the numbers of discs on each side of 0, and by Sylvester's law of inertia these are the inertia of H.

If either step fails on the whole cell, we split P into 2^m and then 4^m sub-cells, tighten E on each with (1), and accept only if every sub-cell yields a certificate with the same u. Nothing in this procedure relies on the floating-point quantities being accurate; inaccurate H or Q only make a test fail.

3.4 Atlas

Starting from P_0, cells are processed breadth first. A cell is certified when Algorithm 1 completes and every equilibrium has an inertia certificate; otherwise it is bisected along its relatively widest free parameter until a depth limit or a time budget is reached. Cells that contain a bifurcation (a fold or a Hopf point, where an equilibrium is non-hyperbolic for some p) cannot be certified at any depth; they are reported as uncertified.

Certified and heuristic outputs. The atlas reports only certified cells as certified. The repository also keeps a fast heuristic mode with a four-stage stability cascade (interval Routh–Hurwitz; Bauer–Fike bounds with a floating-point eigenvector condition number; evaluation on a 10^{-8}-neighbourhood of the cell midpoint; center-manifold rules). Only the first stage is a proof for the whole cell; Section 5.5 measures the others, and every label they produce is marked heuristic.

4 Implementation

PhaseCartographer is written in Python; numerical kernels use NumPy and SciPy, and models are SymPy expressions.

Interval arithmetic. The certified engine's scalar and vectorised (NumPy) kernels round every IEEE operation outward by one ulp with `nextafter`, which is valid because $+, -, \times, /, \sqrt{}$ are correctly rounded; integer powers use repeated outward-rounded multiplication. The general-purpose `Interval` class uses TwoSum error signs for addition and FMA residual signs for multiplication, division and square root, with exact binary-rational comparisons when a tiny residual can underflow and outward finite-input overflow bounds. Exponential and logarithmic endpoints use 100-bit mpmath interval operations and outward conversion to binary64, in both scalar and array evaluators. Both are tested against exact rational arithmetic (Section 5). Unbounded endpoints enclose finite real values, and zero times an overflowed endpoint encloses zero. Invalid interval endpoints are refused or widened to the whole real line before they can support exclusion. State and parameter boxes have finite ordered bounds; zero-width state domains remain unresolved. Exact rational constants in a model are enclosed, not rounded. Right-hand sides are compiled to straight-line interval code after common-subexpression elimination.

Reducing dependency. For polynomial systems the enumeration runs on $g = R_0 f$, where $R_0 \approx J(z_0, p_0)^{-1}$ has been proven nonsingular ($\|I - R_0 J(z_0, p_0)\|_\infty < 1$ in interval arithmetic) and the products are expanded symbolically, so that monomials cancel before interval evaluation. The zeros of g and f coincide; inertia is always computed from f.

SBML and conservation laws. SBML files are read with libSBML; function definitions and assignment rules are expanded, and identifiers are parsed as plain symbols. Conservation laws are computed exactly from the left null space of the stoichiometric matrix; one species per law is eliminated using the totals implied by the initial condition, and non-negativity of the eliminated species becomes an admissibility constraint $g_j \geq 0$. State bounds are the maxima of each species over the stoichiometric compatibility class (a linear program). For exclusion tests only, eliminated species are evaluated as $[\max(0, \underline{g_j}), \overline{g_j}]$, which is valid on admissible points and removes spurious poles outside the admissible set.

Interfaces. `python -m phase_cartographer certify MODEL` builds an atlas for a built-in model, a BioModels entry or a YAML/SBML file and writes a JSON file that records, for every cell, the equilibrium enclosures and their inertia certificates. Exporters write SMT-LIB2 queries and CAPD task files for independent re-checking.

5 Evaluation

5.1 Research questions and setup

RQ1
Are the certificates correct? (floating-point spot checks; independent re-verification)

RQ2
How much of the parameter space is certified, and at what cost?

RQ3
How do heuristic stability labels compare with certified ones?

RQ4
How do certified atlases compare with DSGRN's switching-limit answer?

RQ5
How does the cost of a complete count grow with the state dimension?

Models. Eighteen textbook models with 2–3 states and 1–2 swept parameters (Table 1), and the BioModels entries [19] Goldbeter's mitotic oscillator [10] (BIOMD0000000003, 3 states, sweep $v_i \in [0.005, 0.05]$, $k_d \in [0.005, 0.02]$), Tyson's cell-cycle model [25] (BIOMD0000000005, 6 species reduced to 5, sweep $k_6 \in [0.1, 10]$), Kholodenko's MAPK cascade with negative feedback [17] (BIOMD0000000010, 8 species reduced to 5, sweep $V_1 \in [0.5, 3]$) and the Markevich–Hoek–Kholodenko dual-phosphorylation switch [20] (BIOMD0000000026, 11 species reduced to 8; used for the scaling study). For scaling we also use the Huang–Ferrell MAPK cascade [11] (BIOMD0000000009, 22 species reduced to 15) and synthetic n-gene repression rings. SBML files and SHA-256 sums are in `data/biomodels`.

Protocol. Atlases were built on Modal cloud containers with one CPU core each (Python 3.13), with a depth limit of 12 bisections for the textbook models and Tyson's model and 10 for Goldbeter's and Kholodenko's models (the latter two split over 16 containers each), and a time budget per model; all settings and raw outputs are in `results/atlas`. Inside each certified cell we drew two random parameter points and compared the certified signature with the equilibria found by multistart Newton (100 starts) and their floating-point eigenvalues; differences were re-tested with a denser search (3000 random starts plus a grid of starts).

5.2 RQ1: correctness

Spot checks. Across all certified cells, 22070 random parameter points were checked; at none of them did the floating-point equilibria and eigenvalues differ from the certificate. In 14 of the first-pass checks (two models whose state box contains the origin) the 100-start Newton reference had missed the equilibrium at the origin; the denser search found it in all 14 cases.

Independent re-verification. For up to 12 random certified cells per model we exported three kinds of claims as SMT-LIB2 queries with exact rational constants: *exclusion* (no equilibrium in D outside the enclosures, for any $p \in P$), *stability* (no point of $P \times E_i$ with $f = 0$ violates the Routh–Hurwitz conditions; $n \leq 3$) and *saddle* ($\det J < 0$ on $P \times E_i$, $n = 2$). Every query is unsatisfiable if the certificate is correct. With a 30 s limit, Z3 answered `unsat` on 209/233 exclusion, 159/171 stability and 27/39 saddle queries and returned *unknown* on the rest; it never returned `sat`. dReal answered `unsat` on 168/233, 171/171 and 39/39 queries; its 63 δ-sat answers are inconclusive by design (the equality $f = 0$ is relaxed by $\delta = 10^{-3}$, which admits points next to the enclosures). Together, 431 of the 443 claims were proven by at least one solver. CAPD 6.1.0 (commit `2f06098`), using its own interval arithmetic and automatic differentiation, re-proved existence and uniqueness of a zero in a box around the enclosure for 408 of 577 certified equilibria, and re-proved the certified stability type (Routh–Hurwitz/Liénard–Chipart on the interval characteristic polynomial, or $\det J < 0$ for planar saddles) for 468 of the 514 equilibria where such a test applies. CAPD's plain first-order Krawczyk test lacks the parameter mean-value form and the symbolic preconditioning of Section 4, which may contribute to undecided cases; no controlled ablation establishes this explanation. Fuzz tests of the directed-rounding primitives ($2 \cdot 10^4$ random operand pairs) and of the scalar and vectorised interval kernels ($4 \cdot 10^3$ random interval pairs each) against exact rational arithmetic found no invalid bound. A prospectively fixed scalar control study evaluates $f(x, p) = x^2 - p$ on positive, negative and zero-crossing parameter cells, a rational feedback equation, and roots on state-domain boundaries. A followup adds a narrow rational cell and $f(x, p) = \exp(x) - p$. Of nine cells, four receive certificates with analytically correct counts and inertia; the other five remain uncertified. None receives an incorrect certificate. Separate exact-rational arithmetic controls exercise signed subnormal multiplication and finite-input overflow, while 13 transcendental endpoint cases agree with a 150-digit reference. A further independent exact-rational oracle exercises 29,633 arithmetic cases, including finite inputs whose intermediate reciprocal overflows; all bounds enclose the reference values. An overflowed exponential multiplied by a zero state is also required to remain uncertified rather than receive a false zero-root count. These finite controls test implementation boundaries rather than establish a theorem for all inputs. Plans and raw executions are in `audit/`.

5.3 Fresh analytic comparison and extended-bound attacks

A frozen comparison uses $D = [-2, 2]$, $f = x - p$ and $f = x^2 - p$, and nine declared parameter cells per family. Both scalar and vector backends receive 100 tiles and ten Newton starts. The independent family-specific solver uses the exact formulas $x = p$ and $x = \pm\sqrt{p}$, with signs of 1 and $2x$ for inertia; exact SymPy polynomial root counts at each cell's endpoints and midpoint are retained as a separate diagnostic. Those point diagnostics alone do not establish uniformity, which follows from the analytic formulas. Of 36 backend–cell runs, 22 certify correctly and 14 remain unknown, with no incorrect certificate. Thirty runs have a uniform hyperbolic answer; six fold runs cannot have one because the cell includes $p = 0$. Eight feasible runs remain unknown: four affine boundary runs and four positive fold runs. The vector backend certifies two positive fold cells that the scalar backend leaves unknown. Unknowns remain in the denominator. This demonstrates conservative finite behaviour and a completeness gap, not superiority over Ibex or CAPD; the analytic comparator is restricted to these two families. A biologically consequential multivariable decision and a fair complete-solver benchmark remain open. A fresh seeded extended-bound study tests compositions $(A + B)C$ and A/B using independently calculated exact rational answers for selected finite members. Of 8,000 attempts across both interval backends, 7,009 enclose the exact answer and 991 raise retained domain exceptions; no invalid bound is observed. The 17 targeted overflow/NaN regressions pass. Test scripts, the prospective plan, every unknown, every exception

Model	n	m	Kinetics	Cells	Cert.	Cert. vol.	Certified regimes (volume share)
toggle_switch	2	2	hill	1780	298	63.8%	monostable 63.0%, bistable 0.8%
mutual_activation_switch	2	2	hill	3814	2239	61.5%	bistable 50.8%, one stable of 2 10.8%
michaelis_menten_switch	2	2	rational	2490	1665	79.9%	one stable of 2 73.2%, unique unstable 4.9%, monostable 1.7%
erk_bistable	2	2	rational	59	59	100.0%	monostable 100.0%
apoptosis_switch	2	2	hill	592	496	97.7%	one stable of 2 84.7%, unique unstable 12.9%
mapk_cascade	2	2	polynomial	46	46	100.0%	monostable 100.0%
brusselator	2	2	polynomial	2347	1414	77.2%	monostable 55.1%, unique unstable 22.1%
selkov	2	2	polynomial	1349	798	86.5%	unique unstable 58.3%, monostable 28.3%
fitzhugh_nagumo	2	2	polynomial	779	546	94.3%	unique unstable 61.8%, monostable 32.5%
van_der_pol	2	2	polynomial	190	126	98.4%	unique unstable 98.4%
lotka_volterra	2	2	polynomial	4096	0	0.0%	–
tyson_cell_cycle	2	2	polynomial	72	72	100.0%	monostable 100.0%
p53_mdm2	2	2	rational	898	898	100.0%	monostable 100.0%
calcium_oscillator	2	2	polynomial	365	301	98.4%	monostable 88.2%, no equilibrium in D 10.3%
circadian_clock	2	2	hill	834	834	100.0%	monostable 100.0%
repressilator	3	1	hill	5	5	100.0%	monostable 100.0%
goodwin	3	2	hill	223	223	100.0%	monostable 100.0%
genetic_oscillator	3	2	hill	293	293	100.0%	monostable 100.0%
goldbeter1991 (BIOMD0000000003)	3	2	SBML	1014	77	8.5%	unique unstable 8.5%
tyson1991 (BIOMD0000000005)	5	1	SBML	221	183	99.1%	monostable 88.1%, unique unstable 11.0%
kholodenko2000 (BIOMD0000000010)	5	1	SBML	706	462	76.2%	unique unstable 76.2%

Table 1: Certified atlases. n: state variables (after conservation-law reduction); m: swept parameters; Cells: leaf cells; Cert. vol.: share of the parameter box covered by certified cells. Regime names count stable equilibria in D ("one stable of 2": one stable equilibrium and one saddle; "unique unstable": a single equilibrium with an unstable direction).

and raw results are in `audit/wave2/`. These finite probes are not exhaustive proofs.

Historical provenance of the following evaluation. The atlas, scaling, heuristic and external-checker numbers below describe stored earlier runs. Archive reanalysis counts 21 models, 22,173 cells and 11,035 records labelled certified. The full atlas and the CAPD/Z3/dReal batches have not been independently rerun with the repaired implementation. Their agreement is historical evidence, not post-repair replication or a validation of every archived label. Reported timings likewise belong to those earlier configurations.

5.4 RQ2: coverage and cost

Table 1 lists the atlases. The whole parameter box is certified for 8 models. Tyson's five-state reduction is certified on 99.1% of its parameter range, split into 88.1% monostable and 11.0% unique unstable. Kholodenko's five-state reduction is certified on 76.2%, and every certified cell has a unique unstable equilibrium, consistent with the oscillations reported for this model. Lotka–Volterra is never certified, which is the correct outcome, because its interior equilibrium is a center for every parameter value and therefore not hyperbolic. The toggle switch is certified on 63.8% of its box, but only 0.8% is certified bistable. The bistable wedge is thin and its symmetric saddle is close to singular, so cells there must be much smaller than our depth limit allows (Section 6). In Goldbeter's model (8.5%) the uncertified cells fail at the first step, because the floating-point equilibrium could not be enclosed in a Krawczyk box valid for the whole cell at the cell sizes reached within the depth limit. Single-core wall time per atlas ranged from 1 s (repressilator) to 4837 s summed over shards (Kholodenko).

5.5 RQ3: heuristic versus certified stability labels

We ran the four-stage cascade of the heuristic mode on every equilibrium enclosure of every atlas cell (up to 250 cells per model, including uncertified cells in which some equilibria were isolated) and

compared its label with floating-point eigenvalues at four random parameters of the cell. Stage 1 (interval Routh–Hurwitz on the whole box) labelled 3967 equilibria with no error, and agreed with the certified inertia on all 3799 for which both exist. Stage 3 (midpoint evaluation) labelled 201 equilibria; for 35 of them the label was wrong at some sampled parameter, and every one of these lies in a cell whose equilibrium changes stability inside the cell, where no single label is correct. The certified inertia test produced a label for only 1 of the 201 equilibria that reached Stage 3; the remaining ones are exactly the cases the certificate refuses. Stage 2 never fired, and Stage 4 handled the 250 Lotka–Volterra centers without assigning a stability label. The heuristic stages therefore add labels only where the certified test declines, and roughly one in six of those labels is wrong somewhere in its cell.

5.6 RQ4: comparison with DSGRN

DSGRN describes the steep-switching limit of the same networks. Mapping each certified cell of the toggle switch (Hill coefficient 2) to its DSGRN parameter node, DSGRN predicts two stable fixed points everywhere in $[2, 4]^2$, while the certificates show a single stable equilibrium in 266 of the 298 certified cells. For the three-gene repressilator, DSGRN reports a full cycle (an oscillation-capable Morse node without stable fixed point), while all 5 certified cells have a unique stable equilibrium; for Hill coefficient 2 and equal decay rates the ring cannot lose stability, as the certificate states. DSGRN answers combinatorial questions in milliseconds for whole networks; PhaseCartographer answers the quantitative question for the given kinetics. The two are complementary, and the disagreement quantifies how far the moderate-Hill models are from the switching limit.

5.7 RQ5: scaling with dimension

Table 2 shows that cost depends on structure as much as on dimension. Repression rings are certified up to 16 genes (1.2 million tiles, under 10 minutes); the 20-gene ring was not completed within 10^6 tiles. The eight-state Markevich reduction is certified at its nominal parameters (three equilibria, two of them stable, which is the bistable switch reported in [20]) in 33 s, but a $\pm 0.2\%$ box is not completed within $4 \cdot 10^6$ tiles because two of its equilibria could not be isolated uniformly on that box. The 15-variable Huang–Ferrell reduction is not completed at 10^6 tiles, with about 6% of the state box unresolved.

6 Discussion and limitations

Scope of the certificate. The certificate is about equilibria in a user-chosen state box and their linear stability. It does not certify periodic orbits, basins of attraction, or global behaviour outside D; for SBML models the count refers to the stoichiometric compatibility class fixed by the initial condition. Choosing D is part of the problem statement; equilibria outside D are not counted.

Bifurcations and coverage. Cells that contain a fold or Hopf point cannot be certified, so coverage is limited by how close the depth limit lets cells approach bifurcation sets. Near-singular Jacobians (for example the symmetric saddle of the toggle switch close to its pitchfork) require very small cells, because the Krawczyk test is first order in the parameter width. This is visible in the low certified share of the toggle switch's bistable wedge.

Model	n	box	Status	Tiles	Time (s)	Signature
goldbeter1991	3	point	certified	2,021	0	[2]
goldbeter1991	3	±0.2%	certified	2,029	0	[2]
ring3	3	point	certified	77	0	[0]
ring3	3	±0.2%	certified	85	0	[0]
ring4	4	point	certified	283	0	[0, 0, 1]
ring4	4	±0.2%	certified	315	0	[0, 0, 1]
kholodenko2000	5	point	certified	242,511	9	[2]
kholodenko2000	5	±0.2%	certified	243,619	13	[2]
ring5	5	point	certified	413	0	[2]
ring5	5	±0.2%	certified	435	0	[2]
tyson1991	5	point	certified	481	1	[2]
tyson1991	5	±0.2%	certified	5	1	[2]
ring6	6	point	certified	1,095	0	[0, 0, 1]
ring6	6	±0.2%	certified	1,127	1	[0, 0, 1]
ring7	7	point	certified	1,903	1	[2]
ring7	7	±0.2%	certified	1,921	1	[2]
markevich2004	8	point	certified	254,521	33	[0, 0, 1]
markevich2004	8	±0.2%	incomplete (0.042% of D open)	4,014,199	726	–
ring8	8	point	certified	4,395	1	[0, 0, 1]
ring8	8	±0.2%	certified	4,455	1	[0, 0, 1]
ring10	10	point	certified	17,757	2	[0, 0, 3]
ring10	10	±0.2%	certified	17,933	4	[0, 0, 3]
ring12	12	point	certified	72,495	12	[0, 0, 3]
ring12	12	±0.2%	certified	72,925	16	[0, 0, 3]
huang_ferrell1996	15	point	incomplete (5.766% of D open)	1,012,397	813	–
huang_ferrell1996	15	±0.2%	incomplete (5.787% of D open)	1,013,785	1289	–
ring16	16	point	certified	1,205,389	447	[0, 0, 3]
ring16	16	±0.2%	certified	1,214,559	497	[0, 0, 3]
ring20	20	point	incomplete (0.059% of D open)	1,008,499	621	–
ring20	20	±0.2%	incomplete (0.060% of D open)	1,008,471	660	–

Table 2: Complete equilibrium count at the nominal parameter (point) and on a ±0.2% parameter box, with tile budgets $10^4, 10^5, 10^6, 4 \cdot 10^6$ tried in turn (last run shown). "x% of D open": share of the state box still unresolved when the budget ran out.

Dimension. The cost of a complete count grows quickly with the dimension of the state box, and structure matters. A 16-gene ring is certified in minutes, while the 20-gene ring and the 15-variable Huang–Ferrell reduction were not completed within 10^6 tiles. The current implementation is single-threaded Python; the branch-and-prune loop is vectorised but not compiled.

Floating-point assumptions. Directed rounding assumes IEEE-754 round-to-nearest for $+, -, \times, /, \sqrt{}$. Exponentials and logarithms rely on the mpmath interval backend. The atlas subjects use polynomial or rational expressions; transcendental functions are exercised by separate analytical controls.

7 Conclusion

The implementation assembles established interval zero-solving and inertia arguments into parameter-cell atlases. Fresh finite controls found no incorrect certificate, while the analytic

comparison exposes avoidable unknowns and backend differences. The strongest defensible result is an audited implementation checkpoint. Universal soundness of the software, post-repair biological atlas replication, a fair comparison against mature complete solvers, and a consequential biological decision remain unresolved.

Availability. Code, data, raw results and the commands that produce every number in this paper are in the repository (`REPRODUCE.md`).

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